

1 Latent Representation Probing for Toxic Comment Detection

1.1 Stakeholder Engagement Plan

1.1.1 Identified Stakeholders

Platform operations and trust-and-safety teams. These stakeholders need a reliable, low-latency service that can integrate with moderation queues, flagging systems, and escalation policies.

End users and digital communities. Users want online spaces that are safer and less abusive without unfairly suppressing benign speech, disagreement, humor, or identity-related discussion.

Developers and AI engineers. Technical teams need interpretable interfaces, reproducible notebooks, documented preprocessing, stable model versions, and actionable metrics such as precision, recall, F1 score, and ROC-AUC.

Moderators and policy specialists. Human reviewers need understandable decisions, confidence information, override controls, and a clear process for reporting systematic errors.

1.1.2 Engagement and Deployment Workflow

1.1.2.1 1. Establish requirements

Agree on the operational definition of toxicity, the content categories in scope, the acceptable trade-off between missed harms and over-moderation, and the actions associated with each confidence band.

1.1.2.2 2. Validate with representative cases

Review examples from multiple communities, languages, dialects, and severity levels. Include benign counterexamples that contain profanity or contentious topics but are not intended as abuse.

1.1.2.3 3. Create feedback loops

Use user reports, moderator overrides, appeals, and randomly sampled audits to identify errors. Feed these findings into evaluation and threshold review rather than automatically retraining without governance.

1.1.2.4 4. Provide transparency and interpretability

Document the source dataset, representation layer, pooling strategy, probe type, threshold policy, and known limitations. Activation analysis and representative examples can help explain patterns, but should not be treated as proof of causal reasoning.

1.1.2.5 5. Define service objectives

The supplied outline proposes a target of less than 50 ms classification time per post. This is a deployment goal, not a measured result; benchmark it on the target hardware and traffic profile before committing to the SLO.

1.1.3 Governance Questions

Who can change thresholds? How are policy changes versioned? What evidence is required before automatic action is enabled? How are appeals handled? What retention and privacy rules apply to user content and model logs? These questions should be answered before production rollout.

1.1.4 Success Measures

Track precision and recall by category and subgroup where appropriate, calibration, false-positive and false-negative rates, latency, throughput, moderator override rates, appeal outcomes, and drift over time. Review these measures with both technical and non-technical stakeholders.

Prepared from the supplied project outline; the proposed latency target is explicitly labeled as a goal pending measurement.